One shared proxy implementation

Native C/C++ modules for MyScoutee and MyScoutee Template. The application image inherits this nginx base; both projects pin the same implementation as a Git submodule.

What runs at the edge

Capability	Behavior
Coordinate eligibility	S2 spherical lookup against the complete bundled GeoJSON snapshot. Polygon, MultiPolygon, holes and partition specificity. No per-request backend call.
IP policy	Individual IPv4/IPv6 response messages. Native shared-memory cache and background refresh; updates require no nginx restart or reload.
Scanner defense	100 requests to unknown application paths within 600 seconds create a persistent IP ban with bounded path evidence.
Rate limiting	Native nginx limits: 30 requests/s, burst 120; unknown paths: 2/s, burst 10. Excess requests return 429.
Availability	Authenticated backend leases expire after 20 seconds; public JSON and SSE. Kubernetes reads EndpointSlice snapshots over verified TLS.
Shared behavior	Security headers, transport policy, SSI switches, browser diagnostic endpoint and async browser recorder.

The landing, demo and public content stay available

Coordinate eligibility belongs to the existing /api/auth/eligibility contract. It controls login eligibility rather than applying a geographic IP ban to the site. Backend authentication checks remain authoritative. AUTH_REGION_SECURITY_ENABLED=false disables this gate.

Build-time inputs

The application supplies its countryPartitions.json and an endpoint inventory generated from its Spring, Angular and nginx routes. Literal paths use nginx’s native hash map. The image contains no Go, Python or Lua runtime helper. File refresh, persistence and Kubernetes reads run on a native background thread.

Operate and verify

State	Container location
Operator IP → message list	/etc/nginx/myscoutee-operator/ nginx-blocked-ips.json
Automatic bans and findings	/var/lib/nginx/myscoutee/ scanner-blocked-ips.json
Basic traffic checkpoint	/var/lib/nginx/myscoutee/ traffic-summary.json

Persistent state and removal

Mount the operator directory read-only and the state directory on persistent storage writable by nginx UID 101. Valid policy changes refresh in the background; invalid operator updates retain the last valid cache. Manual messages take precedence. Remove an operator entry or delete an automatic-ban entry from the findings file to clear that ban. Findings are never automatically published to GitHub.

A quick traffic summary, without browser debug

Daily counters show source attribution, estimated sessions, HTML/application requests, demo, login, articles, further activity and no further HTTP activity. Read the current summary inside the container at http://127.0.0.1:9081/internal/traffic-summary. A file checkpoint is written every 30 seconds.

An Instagram browser does not by itself prove a paid click. Lack of follow-up HTTP activity does not prove lack of clicks or reading. Optional detailed browser diagnostics remain separate: when enabled, nginx SSI inserts the async script into served HTML; otherwise the script tag is absent.

Focused container checks

Suite	Coverage
edge-test.py	Shared workers, hot updates, custom IPv4/IPv6 messages, 99/100 threshold, persistent replacement/unban, readiness/SSE, SSI and Meta metrics.
kubernetes-test.py	Local TLS/token validation, ready/terminating endpoints and rejection of incomplete snapshots.
native-test.py	Polygon holes, islands, specificity, invalid coordinates, in-memory lookup; optional full-seed cities and matched load comparison.
Local tools / browser tests	Traffic inference regressions and browser lifecycle/interaction diagnostics. These scripts are not deployed in the runtime image.

Bounds

One state cache holds up to 8,192 scanner/operator IP entries, 16,384 active visit slots and 32 readiness services with 128 instances each. Separate proxy instances have separate in-memory state. Local container measurements are not a production throughput guarantee.

Source and exact commands: github.com/fssrepository/myscoutee-nginx